Module 4: Supervised Learning
Submodule: 4.1 Linear Regression
Lecture: Probabilistic Interpretation of Linear Regression

XCS229 Machine Learning

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Contents

I. Introduction

In this lecture, we discuss regression, laying the foundation for our upcoming sessions. This discussion will lead us to explore a larger unification of machine learning models, a significant development in the field about 20 years ago. We will revisit linear regression and provide a probabilistic interpretation for it. This probabilistic perspective is instrumental because it connects various important models in machine learning.

II. Linear Regression

Linear regression involves determining a linear relationship between a set of input training examples $x^{(i)}$ and corresponding outputs $y^{(i)}$.

Given a set of points $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \ldots, (x^{(n)}, y^{(n)})\}$.

Where the feature vectors $x^{(i)}$ exist in $\mathbb{R}^{d+1}$ for $i = 1, 2, \cdots, n$, including a bias term. $y^{(i)} \in \mathbb{R}$, $i = 1, 2, \cdots, n$. The goal is to find the parameter vector $\theta \in \mathbb{R}^{d+1}$ that minimizes the least squares cost function:

$$\theta = \underset{\theta}{\operatorname{argmin}} \sum_{i=1}^{n} \left(y^{(i)} - h_\theta \left(x^{(i)} \right) \right)^2$$

Here, the hypothesis function is defined as $h_\theta(x) = \theta^T x$.

III. Probabilistic Interpretation

In this section, we will discuss how to interpret linear regression from a probabilistic viewpoint.

To model the relationship between input features $x^{(i)}$ and output $y^{(i)}$, we assume:

$$y^{(i)} = \theta^T x^{(i)} + \epsilon^{(i)} \tag{1}$$

where $\epsilon^{(i)}$ is an error term that captures either unmodeled effects (such as if there are some features very pertinent to predicting housing price, but that we left out of the regression) or random noise.

Properties of $\epsilon^{(i)}$:

- We assume that the errors ($\epsilon^{(i)}$) are **independently and identically distributed** (**IID**) random variables that follow a **Gaussian distribution** (also called a Normal distribution).
- The errors are presumed to be IID, so:
  - Unbiased: $\mathbb{E}[\epsilon^{(i)}] = 0$.
  - Independent: $\epsilon^{(i)}$ is independent of $\epsilon^{(j)}$ for $i \neq j$, i.e

    $$\mathbb{E}[\epsilon^{(i)}\epsilon^{(j)}] = \mathbb{E}[\epsilon^{(i)}] \cdot \mathbb{E}[\epsilon^{(j)}].$$

  - Gaussian: Follow a Gaussian distribution with $\mu = 0$ and some variance σ^2:

    $$\mathbb{E}\left[\left(\epsilon^{(i)} \right)^2 \right] = \sigma^2,$$

    We can write this as:

    $$\epsilon^{(i)} \sim \mathcal{N}(0, \sigma^2).$$

    and the density of $\epsilon^{(i)}$ is given by

    $$p\left(\epsilon^{(i)}\right) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{\left(\epsilon^{(i)}\right)^2}{2\sigma^2} \right).$$

From equation (1) we get

$$p\left(y^{(i)}|x^{(i)};\theta\right) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{\left(y^{(i)} - \theta^T x^{(i)}\right)^2}{2\sigma^2}\right). \tag{2}$$

Note:

- The notation $p\left(y^{(i)}|x^{(i)};\theta\right)$ indicates that this is the distribution of $y^{(i)}$ given $x^{(i)}$ and parameterized by θ.
- We should not condition on θ in $p\left(y^{(i)}|x^{(i)};\theta\right)$ since θ is not a random variable.
- We can also write the distribution of $y^{(i)}$ as

$$y^{(i)} \mid x^{(i)};\theta \sim \mathcal{N}\left(\theta^T x^{(i)}, \sigma^2\right).$$

where picking a θ effectively picks a distribution. σ can be measured by looking at the noise in the outputs y.

IV. Gaussian Distribution

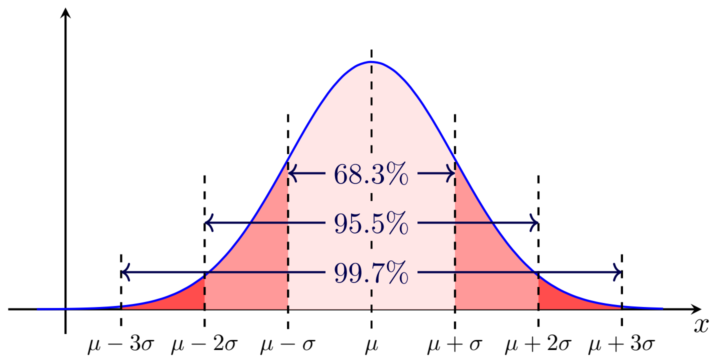
In this section, we delve deeper into the properties and implications of the Gaussian distribution.

The probability density function (PDF) of a Gaussian distribution is defined as:

$$f(y;\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y-\mu)^2}{2\sigma^2}}$$

This distribution is characterized by two parameters:

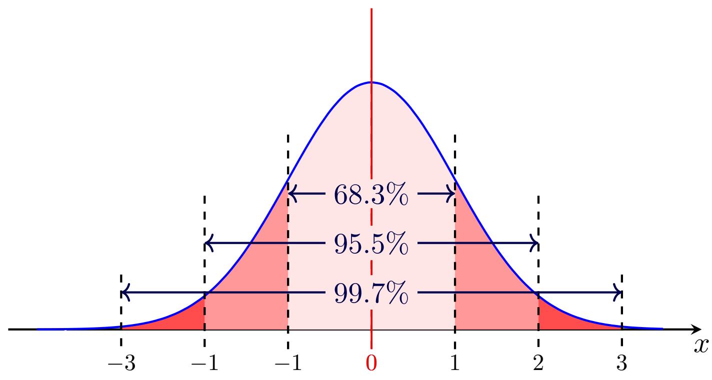
- Mean (μ).
- Variance (σ^2).



Key properties include:

- Symmetry about the mean.
- Total area under the curve equals 1.
- Approximately 68% of the values lie within one standard deviation (σ), 95% within two and 99.7% within three.

Standard normal distribution occurs when a normal random variable has a mean equal to zero ($\mu = 0$) and a standard deviation equal to one ($\sigma = 1$).



V. Likelihood Function

In this section, we will define the likelihood function and its significance.

Given X (the design matrix, which contains all the $x^{(i)}$s) and θ, what is the distribution of the $y^{(i)}$s? The probability of the data is given by $p(\vec{y}|X;\theta)$. This quantity is typically viewed a function of $\vec{y}$ (and perhaps X) for a fixed value of θ. When we wish to explicitly view this as a function of θ, we will instead call it the **likelihood** function.

The **likelihood** $L(\theta)$ of observing the data X given parameters θ is defined as:

$$L(\theta) = p(\vec{y}|X;\theta).$$

Note: Probability and likelihood are both related to the chance of something happening, but they are used in different contexts. Probability quantifies the likelihood of an event occurring, while likelihood is used to assess the plausibility of a parameter value given observed data. In essence, probability is about the uncertainty of an outcome, and likelihood is about the uncertainty of a parameter.

Using the independence assumption on the $\epsilon^{(i)}$'s (and hence also the $y^{(i)}$s given the $x^{(i)}$s), the likelihood can also be written as:

$$L(\theta) = \prod_{i=1}^{n} p\left(y^{(i)} \mid x^{(i)}; \theta\right)$$

From equation (2) we get

$$\begin{aligned}
L(\theta) &= p(\vec{y}|X;\theta) \\
&= \prod_{i=1}^{n} p\left(y^{(i)} \mid x^{(i)}; \theta\right) \\
&= \prod_{i=1}^{n} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{\left(y^{(i)} - \theta^T x^{(i)}\right)^2}{2\sigma^2}\right)
\end{aligned}$$

The principle of **maximum likelihood** suggests that we should choose the parameter θ in such a way that the likelihood of the data is maximized. In other words, we aim to maximize the likelihood function $L(\theta)$.

Instead of directly maximizing $L(\theta)$, we can maximize any strictly increasing function of $L(\theta)$, which simplifies the derivations. One common transformation is the **log likelihood**, denoted as $\ell(\theta)$. We can express the log likelihood as follows:

$$\begin{aligned}
\ell(\theta) &= \log L(\theta) \\
&= \log \prod_{i=1}^{n} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{\left(y^{(i)} - \theta^T x^{(i)}\right)^2}{2\sigma^2}\right) \\
&= \sum_{i=1}^{n} \log\left(\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{\left(y^{(i)} - \theta^T x^{(i)}\right)^2}{2\sigma^2}\right)\right) \\
&= n \log\left(\frac{1}{\sigma\sqrt{2\pi}}\right) - \frac{1}{2\sigma^2} \sum_{i=1}^{n} \left(y^{(i)} - \theta^T x^{(i)}\right)^2
\end{aligned}$$

Maximizing the log-likelihood function $\ell(\theta)$ is equivalent to *minimizing* the following expression:

$$\frac{1}{2} \sum_{i=1}^{n} \left(y^{(i)} - \theta^T x^{(i)}\right)^2$$

This expression is recognized as the least squares cost function $J(\theta)$. So, by maximizing the log-likelihood, we can recover our previously defined loss function related to least squares.

To summarize, under the probabilistic assumptions made on the data, least squares regression can be interpreted as the **maximum likelihood**

estimation (MLE) of the parameter θ. This provides a natural justification for least squares regression as an MLE method.

However, it is important to note that these probabilistic assumptions are not necessary for least squares to be a rational procedure. There are alternative assumptions that can also justify the least squares approach.

The final estimate of θ did not depend on the value of σ^2. This observation remains true even when σ^2 is unknown.

VI. Summary

- **Linear regression**: can be viewed as a maximum likelihood estimation problem under the assumption that the errors are independent, identically distributed, and follow a Gaussian distribution.

- **The likelihood function**: is based on the product of Gaussian distributions for each data point, and the log likelihood simplifies to a sum of squared residuals.

- **Maximizing the likelihood**: is equivalent to minimizing the sum of squared errors, which is the well-known objective in linear regression. This derivation connects the concept of maximum likelihood with the familiar least-squares method for fitting a linear regression model.